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Collegio di Elettronica, Telecomunicazioni e Fisica

Report for the course Integrated Systems Architecture

Master degree in Electrical Engineering

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CHAPTER 1

Lab 2: digital arithmetic

1.1 FPU model

1.1.1 Simulation

The FPU has been simulated using the values listed in Table 1.1, which correspond to the hexadecimal configuration shown there. The hexadecimal values were not obtained by directly converting the decimal numbers (e.g., converting 7 to 0x0007). Instead, the numbers were first processed through `fp16_num.cpp` to generate their binary `bfloat16` representation, and this binary form was then converted to hexadecimal.

For example, the number 7 in `bfloat16` format is represented as 0100 0000 1110 0000, which corresponds to the hexadecimal value 0x40E0.

<i>a</i>		<i>b</i>		<i>r</i>	
7	0x40E0	-896	0xC460	-6272	0xC5C4
25	0x41C8	25	0x41C8	625	0x441C
128	0x4300	128	0x4300	16384	0x4680
896	0x4460	-112	0xC2E0	-100352	0xC7C4
112	0x42E0	-7	0xC0E0	-784	0xC444
-112	0xC2E0	-25	0xC1C8	2800	0x452F
-896	0xC460	-896	0x4460	-802816	0xC944
-7	0xC0E0	-128	0xC300	896	0x4460
-25	0xC1C8	7	0x40E0	-175	0xC32F
-128	0xC300	112	0x42E0	-14336	0xC660

Table 1.1: Values used for testing the FPU

Figure 1.1 shows a snapshot of the simulation. As it can be observed by simulating the FPU with the numbers shown in Table 1.1, the results are the expected ones.

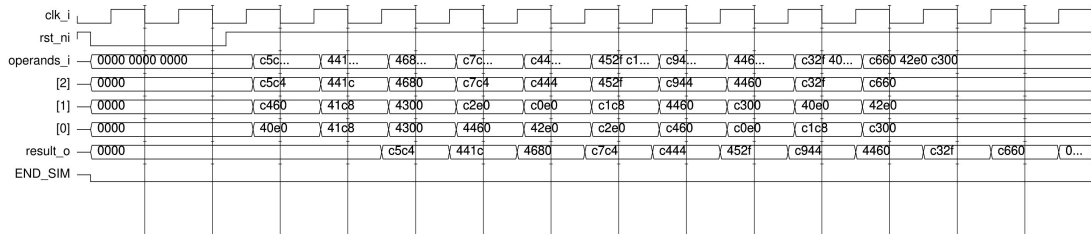


Figure 1.1: Comparison between data from C++ and `fpnew_top` output results

For comparison, we chose to use the output factor D2 (`operands_i[2]`) as the reference, since it corresponds to the results produced by the C++ file `fp16_num.cpp`. The input data are `operands_i[0]` (a) and `operands_i[1]` (b). The output of `fpnew_top` appears on the signal `result_o`, which is delayed by two clock cycles with respect to the first input data. Given this timing,

operands.i[2] and result.o can be directly compared in Figure 1.1. It is clear that the two signals match, confirming that the model is functioning correctly.

1.1.2 Synthesis

In the following figures, there are snapshots of the netlist simulation realized between 0 ns and 90 ns. The signal /tb_fpnew_top/operands_i[2] represents the results obtained through the C++ code, as it's shown it perfectly matches the results shown in signal /tb_fpnew_top/UUT/result_o. Table 1.2 shows the results of the different synthesis methods required in the assignment.

Synthesis strategy	Period	Frequency	Area
compile	$T_s = 4.5 \text{ ns}$	$f_M = 222 \text{ MHz}$	$A = 3674 \mu\text{m}^2$
compile and optimize_registers	$T_s = 4 \text{ ns}$	$f_M = 250 \text{ MHz}$	$A = 3149 \mu\text{m}^2$
compile_ultra	$T_s = 4.3 \text{ ns}$	$f_M = 233 \text{ MHz}$	$A = 2563 \mu\text{m}^2$
CSA multiplier	$T_s = 2.5 \text{ ns}$	$f_M = 400 \text{ MHz}$	$A = 3428 \mu\text{m}^2$
PPARCH multiplier	$T_s = 2.5 \text{ ns}$	$f_M = 400 \text{ MHz}$	$A = 3405 \mu\text{m}^2$

Table 1.2: Simulation results using different synthesis strategies

While the **compile** command performs a standard synthesis that can support the optimization of the design for area, timing and power, the **compile_ultra** command focuses on high performances designs. It adds to the standard command actions such as automatic ungrouping, automatic boundary optimization, automatic datapath extraction and optimization and register retiming. The **optimize_registers** command though, issued after **compile**, performs retiming of sequential cells, a function that **compile_ultra** performs automatically.

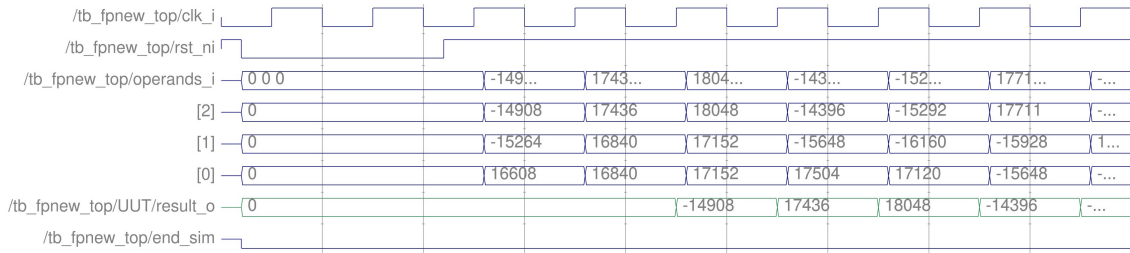


Figure 1.2: Waveform of the netlist obtained through command **compile**

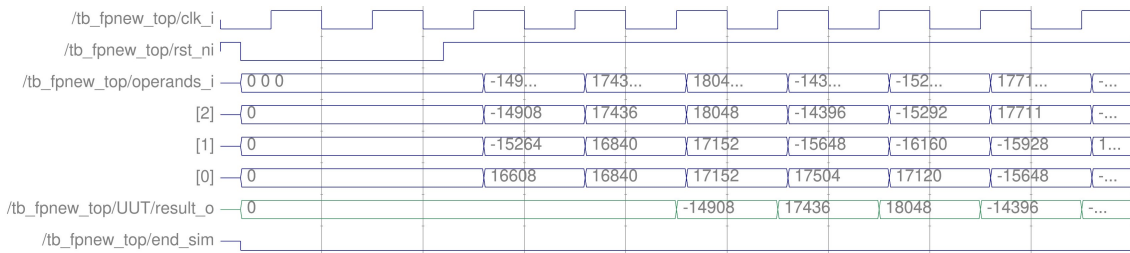


Figure 1.3: Waveform of the netlist obtained through commands **compile** and **optimize_registers**

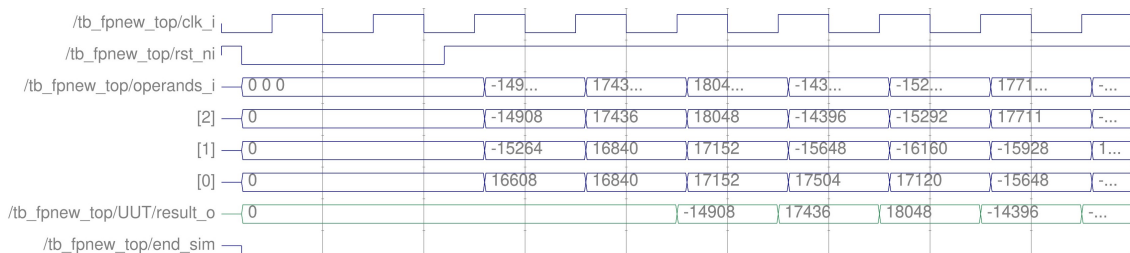


Figure 1.4: Waveform of the netlist obtained through command **compile_ultra**

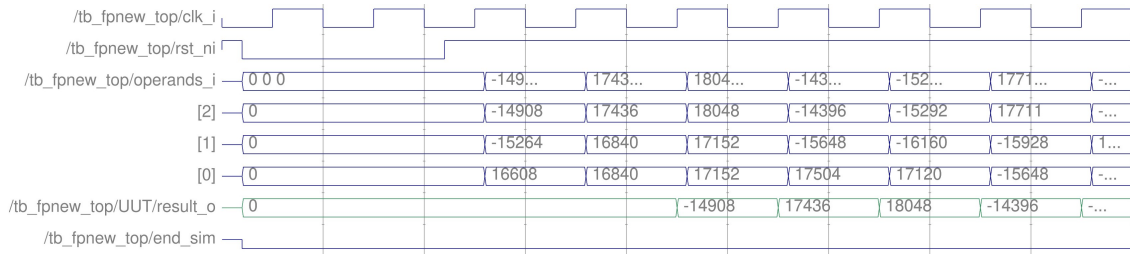


Figure 1.5: Waveform of the netlist obtained through CSA multiplier and commands `compile` and `optimize_registers`

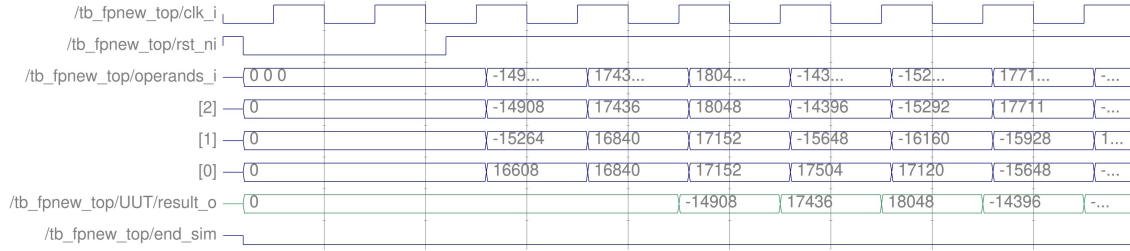


Figure 1.6: Waveform of the netlist obtained through PPARCH multiplier and commands `compile` and `optimize_registers`

1.1.3 Explanations, comparisons and comments

As shown in Figures 1.2–1.6, the waveforms obtained from the simulations are identical. The main differences are summarized in Table 1.2. Depending on the synthesis strategy, different critical paths are produced, leading to different achievable maximum clock frequencies. While more complex synthesis architectures result in shorter periods, this comes at the cost of increased chip area.

1.2 R4-MBE multiplier

The multiplier is implemented using a parallel booth encoder architecture. The "Multiplier" block has to replace the operation at line 326 in `fpnew_fma.sv`, which multiply the value of the 8 bits word `mantissa_a` and `mantissa_b` and assign the result to the 16-bit word `product`. The circuit implementing the R4-MBE is shown in Fig. 1.7. To properly apply the booth encoding approach, the `mantissa_b` was extended by adding a dummy FF at the LSB, and 2 dummy FF at the MSB. 5 instances of the booth encoder were allocated. In the booth encoder, a bit was appended to carry a separated information about the carry bit to inject into the wallace tree. The wallace tree was allocated using the architecture shown in 1.8. Moreover, the figure shows how sign extension has been simplified to avoid unnecessary compressors.

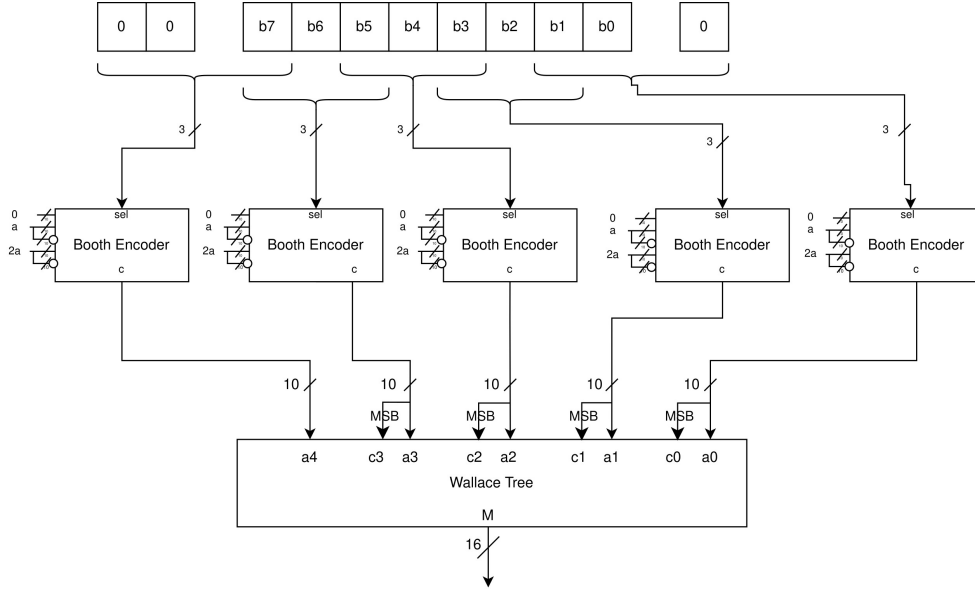


Figure 1.7: Complete Circuit design of R4-MBE

In figure 1.7, the inputs of the Booth Encoders are $0, a, 2a, -a, -2a$. In particular $2a$ is obtained by a left shift of the a operand, $-a$ from the inversion of a without adding 1, $-2a$ from the inversion $2a$ without adding 1. These inputs are on 9 bits.

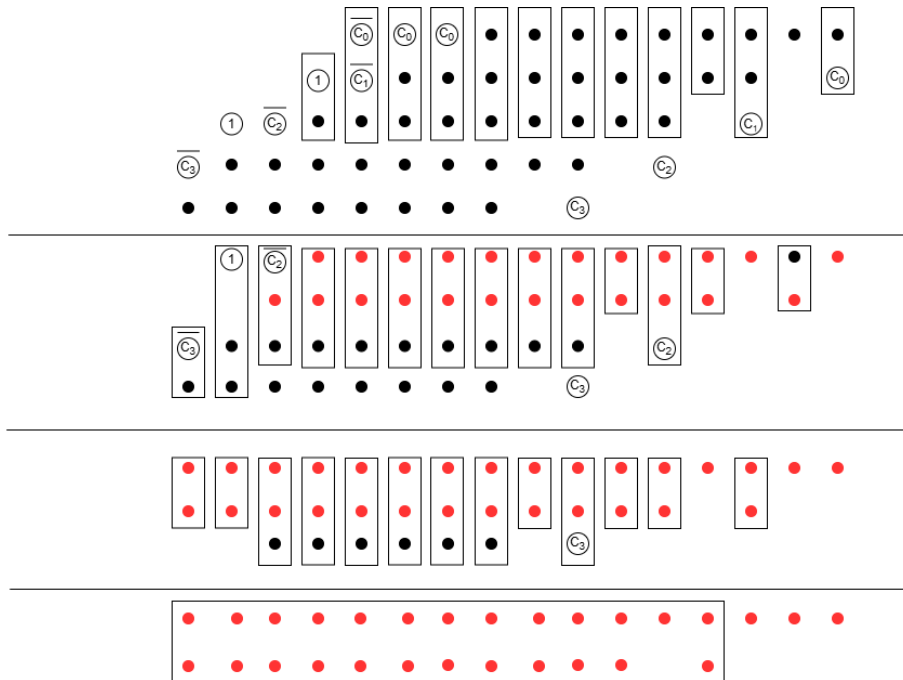


Figure 1.8: Caption design of the Wallace Tree

1.2.1 Simulation

Standalone multiplier To properly simulate the behavior of the **Multiplier** an ad hoc self-checking testbench has been created. The testbench `tb_MantMult` automatically create 200 couples of random 8 bits numbers and feed them sequentially in an interval of 200ns. For every couple of inputs, the testbench evaluate the theoretical product of the inputs and compare the results via a difference. The result of this evaluation are all printed in a file called `outputs.txt`. A snippet of `outputs.txt` is in 1, the complete version is reported in the appendix. A snapshot of the simulation is shown in Figure 1.9.

```
.
.
.
A: 3 | B:144 | M: 432 | Ideal_M: 432 | Error: 0
A:162 | B:182 | M:29484 | Ideal_M:29484 | Error: 0
A: 62 | B: 36 | M: 2232 | Ideal_M: 2232 | Error: 0
A: 98 | B:215 | M:21070 | Ideal_M:21070 | Error: 0
A:103 | B:164 | M:16892 | Ideal_M:16892 | Error: 0
A:136 | B:192 | M:26112 | Ideal_M:26112 | Error: 0
A:233 | B: 55 | M:12815 | Ideal_M:12815 | Error: 0
A:131 | B:174 | M:22794 | Ideal_M:22794 | Error: 0
A: 90 | B:116 | M:10440 | Ideal_M:10440 | Error: 0
A:216 | B:201 | M:43416 | Ideal_M:43416 | Error: 0
A:197 | B:218 | M:42946 | Ideal_M:42946 | Error: 0
.
.
.
```

Code 1: Snippet of output.txt

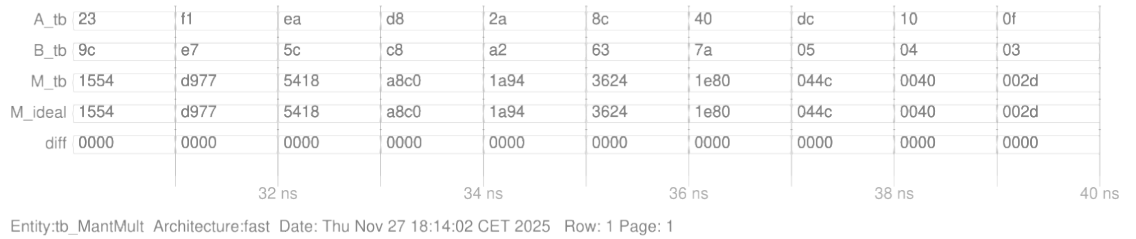


Figure 1.9: Simulation snippet of standalone R4-MBE multiplier

Whole FPU Figure 1.10 shows a snapshot of a simulation for the designed multiplier included in the whole FPU.

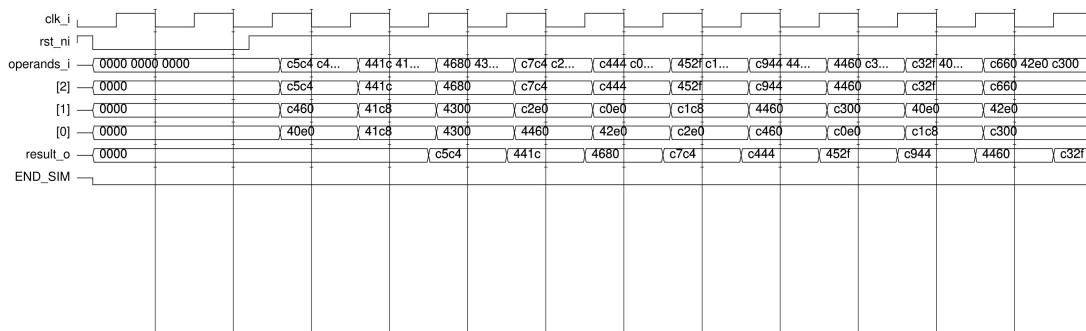


Figure 1.10: Simulation snippet of entire FPU with custom R4-MBE multiplier

From both the standalone multiplier simulation and the full FPU simulation, it is clear that the designs operate correctly. Comparing Figure 1.10 with Figure 1.1, the FPU integrating the custom R4-MBE multiplier exhibits the same output behavior as the FPU using the ideal multiplier.

1.2.2 Synthesis

Synthesis strategy	Period	Frequency	Area
compile	$T_s = 2.7 \text{ ns}$	$f_M = 370 \text{ MHz}$	$A = 3493 \mu\text{m}^2$
compile and optimize_registers	$T_s = 2.2 \text{ ns}$	$f_M = 454 \text{ MHz}$	$A = 2751 \mu\text{m}^2$
compile_ultra	$T_s = 2.5 \text{ ns}$	$f_M = 400 \text{ MHz}$	$A = 2410 \mu\text{m}^2$

Table 1.3: Simulation results using different synthesis strategies with custom R4-MBE multiplier

1.2.3 Explanations, comparisons and comments

Comparing table 1.2 with table 1.3, many improvements can be observed:

- **compile**: Achieves a 66% increase in frequency and a 5% reduction in area.
- **compile and optimize_registers**: Achieves a 13.5% increase in frequency and a 23.7% reduction in area.
- **compile_ultra**: Achieves a 71.6% increase in frequency and a 6.3% reduction in area.

Overall, the characteristics of the FPU improve in both maximum achievable speed and area consumption. This confirms that adopting an R4-MBE multiplier architecture is a strong choice compared to standard multiplier implementations.

Note: The comparison for **compile and optimize_registers** is performed between the synthesis strategy **compile and optimize_registers** in Table 1.3 and the **PPARCH** configuration in Table 1.2. The area comparison for identical synthesis strategies is carried out assuming a clock frequency of 100 MHz.

In all cases, T_s is obtained by attempting to meet a slack condition of 0 ns.

.1 Synthesis strategies

As it was mentioned before, area and resources reports were performed using a clock frequency of 100 MHz to get comparable results, on the other hand timing reports were performed trying to get the minimum slack as to find the maximum frequency for each strategy, so the timing clock frequencies follow the ones listed before.

.1.1 compile

area

```
*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Wed Nov 12 18:22:21 2025
*****

Library(s) Used:

      NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
      NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                1986
Number of nets:                 4241
Number of cells:                2468
Number of combinational cells:  2242
Number of sequential cells:     160
Number of macros/black boxes:   0
Number of buf/inv:              442
Number of references:           15

Combinational area:             2819.068008
Buf/Inv area:                   268.926000
Noncombinational area:          855.190027
Macro/Black Box area:           0.000000
Net Interconnect area:          undefined (Wire load has zero net area)

Total cell area:                3674.258035
Total area:                     undefined
```

timing

```
*****
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Wed Nov 12 18:22:03 2025
*****

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
            [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
            lane_instance.i_fma/inp_pipe_op_q_reg[1][0]
            (rising edge-triggered flip-flop clocked by MY_CLK)
Endpoint:  gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
            active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
            i_fma/mid_pipe_sum_q_reg[1][27]
            (rising edge-triggered flip-flop clocked by MY_CLK)
```

Path Group: MY_CLK

Path Type: max

Des/Clust/Port	Wire Load Model	Library
fpnew_top	5K_hvrat1o_1_1	NangateOpenCellLibrary
Point	Incr	Path
--		
clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/inp_pipe_op_q_reg[1][0]/CK (DFFR_X1)	0.00	0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/inp_pipe_op_q_reg[1][0]/Q (DFFR_X1)	0.10	0.10 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U981/ZN (NOR2_X1)	0.11	0.21 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U824/ZN (INV_X1)	0.06	0.27 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U158/ZN (NAND2_X1)	0.07	0.34 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U822/ZN (OAI21_X1)	0.06	0.40 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_1_root_add_1_root_sub_287/B[1] (fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_add_4_DW01_add_5)	0.00	0.40 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_1_root_add_1_root_sub_287/U1/Z (XOR2_X1)	0.08	0.48 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_1_root_add_1_root_sub_287/U2/Z (XOR2_X1)	0.08	0.56 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_1_root_add_1_root_sub_287/SUM[1] (fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_add_4_DW01_add_5)	0.00	0.56 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_0_root_add_1_root_sub_287/U1_1/S (FA_X1)	0.14	0.69 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U156/ZN (OR2_X1)	0.03	0.73 r

```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/B[1] (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_sub_2)
                                0.00          0.73 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U19/ZN (INV_X1)
                                0.03          0.75 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_1/CO (FA_X1)
                                0.11          0.86 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U4/ZN (NAND2_X1)
                                0.03          0.90 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U6/ZN (NAND3_X1)
                                0.04          0.93 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_3/CO (FA_X1)
                                0.09          1.02 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_4/CO (FA_X1)
                                0.09          1.11 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_5/CO (FA_X1)
                                0.09          1.20 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_6/CO (FA_X1)
                                0.09          1.29 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_7/CO (FA_X1)
                                0.09          1.38 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U2_8/CO (FA_X1)
                                0.09          1.47 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/U12/ZN (XNOR2_X1)
                                0.08          1.56 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_293/DIFF[9] (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_sub_2)
                                0.00          1.56 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U963/ZN (INV_X1)
                                0.04          1.60 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U38/ZN (OAI21_X1)
                                0.05          1.65 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U171/ZN (NAND3_X2)
                                0.05          1.70 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].

```

```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U956/ZN (INV_X1)
                                0.05      1.75 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U955/ZN (NAND2_X1)
                                0.05      1.81 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U39/ZN (OAI22_X1)
                                0.07      1.87 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U887/ZN (OAI22_X1)
                                0.04      1.91 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U154/ZN (AOI211_X1)
                                0.08      1.99 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U806/ZN (INV_X1)
                                0.03      2.02 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U805/ZN (NAND3_X1)
                                0.03      2.05 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U802/ZN (AOI21_X1)
                                0.03      2.08 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U801/ZN (OAI221_X1)
                                0.04      2.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U800/ZN (NOR3_X1)
                                0.04      2.16 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/CI (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_add_5_DW01_add_6
    )
                                0.00      2.16 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U19/ZN (AND2_X1)
                                0.05      2.21 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1/ZN (AND2_X1)
                                0.04      2.25 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_2/CO (FA_X1)
                                0.09      2.34 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_3/CO (FA_X1)
                                0.09      2.43 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_4/CO (FA_X1)
                                0.09      2.52 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].

```

```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_5/CO (FA_X1)
                                0.09      2.61 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_6/CO (FA_X1)
                                0.09      2.70 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_7/CO (FA_X1)
                                0.09      2.79 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_8/CO (FA_X1)
                                0.09      2.88 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_9/CO (FA_X1)
                                0.09      2.97 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_10/CO (FA_X1)
                                0.09      3.06 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_11/CO (FA_X1)
                                0.09      3.16 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_12/CO (FA_X1)
                                0.09      3.25 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_13/CO (FA_X1)
                                0.09      3.34 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_14/CO (FA_X1)
                                0.09      3.43 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_15/CO (FA_X1)
                                0.10      3.53 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U8/ZN (NAND2_X1)
                                0.03      3.56 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U10/ZN (NAND3_X1)
                                0.04      3.60 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U1_17/CO (FA_X1)
                                0.10      3.69 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/U24/Z (XOR2_X1)
                                0.08      3.78 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/add_1_root_add_368_2/SUM[18] (
    ffnw_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_add_5_DW01_add_6
    )
                                0.00      3.78 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].

```

```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/B[18] (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_sub_0)
                                0.00          3.78 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U71/ZN (INV_X1)
                                0.04          3.81 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U14/ZN (AND2_X1)
                                0.05          3.86 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U13/ZN (AND2_X1)
                                0.05          3.91 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U9/ZN (AND2_X1)
                                0.05          3.96 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U12/ZN (AND2_X1)
                                0.05          4.01 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U11/ZN (AND2_X1)
                                0.05          4.05 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U8/ZN (AND2_X1)
                                0.05          4.10 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U3/ZN (AND2_X1)
                                0.05          4.15 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U10/ZN (AND2_X1)
                                0.05          4.20 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U18/ZN (NAND2_X1)
                                0.03          4.23 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/U1/Z (XOR2_X1)
                                0.07          4.30 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/sub_372/DIFF[27] (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_sub_0)
                                0.00          4.30 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U700/ZN (AOI22_X1)
                                0.04          4.35 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U163/ZN (NAND2_X1)
                                0.03          4.38 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/mid_pipe_sum_q_reg[1][27]/D (DFFR_X1)
                                0.01          4.38 f
data arrival time
                                4.38

```

```

clock MY_CLK (rise edge)                4.50      4.50
clock network delay (ideal)              0.00      4.50
clock uncertainty                        -0.07      4.43
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/mid_pipe_sum_q_reg[1][27]/CK (DFFR_X1)          0.00      4.43 r
library setup time                      -0.04      4.39
data required time                      4.39
--
-----

data required time                      4.39
data arrival time                      -4.38
--
-----

slack (MET)                             0.00

```

.1.2 compile and optimize registers

area

```

*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Wed Nov 12 18:42:01 2025
*****

```

Library(s) Used:

```

NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
NangateOpenCellLibrary_typical_ecsm.db)

```

```

Number of ports:                1931
Number of nets:                 3137
Number of cells:                2163
Number of combinational cells:  1728
Number of sequential cells:     190
Number of macros/black boxes:   0
Number of buf/inv:              274
Number of references:           14

Combinational area:             2138.373998
Buf/Inv area:                   148.694001
Noncombinational area:          1010.800033
Macro/Black Box area:           0.000000
Net Interconnect area:          undefined (Wire load has zero net area)

Total cell area:                 3149.174031
Total area:                     undefined

```

timing

```

*****
Report : timing
       -path full
       -delay max
       -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Wed Nov 12 18:41:45 2025
*****

```

Operating Conditions: typical Library: NangateOpenCellLibrary
Wire Load Model Mode: top

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
lane_instance.i_fma/MY_CLK_r_REG33_S2

(rising edge-triggered flip-flop clocked by MY_CLK)

Endpoint: status_o[UF]

(output port clocked by MY_CLK)

Path Group: MY_CLK

Path Type: max

Des/Clust/Port	Wire Load Model	Library
fpnew_top	5K_hvratio_1_1	NangateOpenCellLibrary

Point	Incr	Path
--		
clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/MY_CLK_r_REG33_S2/CK (DFFR_X1)	0.00	0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/MY_CLK_r_REG33_S2/Q (DFFR_X1)	0.12	0.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/in_i[5] (lzc_00000013_1)	0.00	0.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/U13/ZN (NOR3_X1)	0.03	0.15 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/U9/ZN (AND2_X1)	0.05	0.20 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/U44/ZN (NOR2_X1)	0.09	0.29 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/U32/ZN (AOI221_X1)	0.05	0.35 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/U6/ZN (INV_X1)	0.04	0.39 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_lzc/cnt_o[1] (lzc_00000013_1)	0.00	0.39 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U221/ZN (INV_X1)	0.04	0.43 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U48/ZN (NAND2_X1)		

gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U50/ZN (NAND3_X1)	0.03	0.47	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U62/ZN (NAND2_X1)	0.05	0.51	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U64/ZN (NAND3_X1)	0.04	0.55	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U94/ZN (NAND2_X1)	0.05	0.60	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U96/ZN (NAND3_X1)	0.04	0.63	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/sub_1_root_add_512/U2_4/C0 (FA_X1)	0.04	0.67	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U201/ZN (OR2_X1)	0.10	0.77	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U202/ZN (OR2_X1)	0.07	0.84	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U252/ZN (XNOR2_X1)	0.06	0.89	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U286/ZN (NAND3_X1)	0.04	0.94	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U41/ZN (NOR4_X2)	0.04	0.98	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U177/ZN (XNOR2_X1)	0.10	1.07	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U101/ZN (NAND2_X1)	0.07	1.14	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U100/ZN (AND2_X1)	0.03	1.17	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U580/ZN (AOI22_X1)	0.05	1.22	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U579/ZN (INV_X1)	0.06	1.27	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].	0.03	1.30	f

```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U225/ZN (AOI21_X2)
                                0.07      1.37 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U578/ZN (INV_X1)
                                0.04      1.41 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U31/ZN (NAND2_X1)
                                0.11      1.52 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U43/Z (CLKBUF_X1)
                                0.12      1.65 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U539/ZN (OAI221_X1)
                                0.07      1.71 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U53/ZN (AND2_X1)
                                0.05      1.76 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U54/ZN (NOR3_X1)
                                0.09      1.86 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U68/ZN (OAI221_X1)
                                0.06      1.92 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U65/ZN (INV_X1)
                                0.06      1.98 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U231/ZN (NAND3_X2)
                                0.08      2.05 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U67/ZN (INV_X1)
                                0.07      2.12 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U507/ZN (AOI222_X1)
                                0.05      2.17 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U506/ZN (INV_X1)
                                0.04      2.21 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U309/ZN (NAND4_X1)
                                0.04      2.26 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U26/ZN (OAI21_X1)
                                0.15      2.41 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U458/ZN (INV_X1)
                                0.07      2.48 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U316/ZN (OAI211_X1)

```

gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/round_sticky_bits_i[1] (fpnew_rounding_0000000f)	0.07	2.55	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U52/ZN (OR2_X1)	0.00	2.55	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U30/ZN (NAND2_X1)	0.05	2.60	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U54/ZN (OAI211_X1)	0.03	2.63	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U5/ZN (AND2_X1)	0.04	2.67	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U6/ZN (AND2_X1)	0.05	2.72	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U31/ZN (AND2_X1)	0.04	2.76	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U36/ZN (AND2_X1)	0.04	2.80	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U35/ZN (NAND2_X1)	0.05	2.85	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U34/ZN (XNOR2_X1)	0.03	2.88	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/abs_rounded_o[14] (fpnew_rounding_0000000f)	0.07	2.95	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U493/ZN (NAND4_X1)	0.00	2.95	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U492/ZN (NOR2_X1)	0.05	3.00	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U491/ZN (OAI21_X1)	0.05	3.05	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U58/ZN (AND2_X1)	0.04	3.08	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U29/ZN (OR4_X1)	0.05	3.13	f
	0.09	3.22	f

```

gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/U311/ZN (NOR4_X1)
                                0.05          3.28 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/status_o[UF] (
  fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB_)
                                0.00          3.28 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/U38/ZN (AND2_X1)
                                0.05          3.32 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/status_o[UF] (
  fpnew_opgroup_fmt_slice_0_4_00000010_0_00000002_3__logic_Z_1yB_)
                                0.00          3.32 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (
  rr_arb_tree_00000005_1_024242)
                                0.00          3.32 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/U20/ZN (AND2_X1)
                                0.04          3.36 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (
  rr_arb_tree_00000005_1_024242)
                                0.00          3.36 r
gen_operation_groups[0].i_opgroup_block/status_o[UF] (
  fpnew_opgroup_block_0_00000010_0_01_4_155_3__logic_Z_1yB__395949)
                                0.00          3.36 r
i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00          3.36 r
i_arbiter/U18/ZN (AND2_X1)
                                0.04          3.40 r
i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00          3.40 r
status_o[UF] (out)
                                0.02          3.43 r
data arrival time
                                3.43

clock MY_CLK (rise edge)
                                4.00          4.00
clock network delay (ideal)
                                0.00          4.00
clock uncertainty
                                -0.07         3.93
output external delay
                                -0.50         3.43
data required time
                                3.43
--

```

```

-----
data required time
                                3.43
data arrival time
                                -3.43
--

```

```

-----
slack (MET)
                                0.00

```

.1.3 compile_ultra

area

```

*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Wed Nov 12 18:53:38 2025
*****

```

Library(s) Used:

```

NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
NangateOpenCellLibrary_typical_ecsm.db)

```

```

Number of ports:          96
Number of nets:          1810
Number of cells:         1561
Number of combinational cells: 1418
Number of sequential cells:  141
Number of macros/black boxes: 0
Number of buf/inv:        201
Number of references:     41

Combinational area:      1809.864007
Buf/Inv area:            109.858001
Noncombinational area:   752.780024
Macro/Black Box area:    0.000000
Net Interconnect area:   undefined (Wire load has zero net area)

Total cell area:         2562.644031
Total area:              undefined

```

timing

```
*****
```

```
Report : timing
```

```

    -path full
    -delay max
    -max_paths 1

```

```
Design : fpnew_top
```

```
Version: S-2021.06-SP4
```

```
Date   : Wed Nov 12 18:53:10 2025
```

```
*****
```

```
Operating Conditions: typical   Library: NangateOpenCellLibrary
```

```
Wire Load Model Mode: top
```

```

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
             [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
             lane_instance.i_fma/mid_pipe_sum_q_reg[1][9]
             (rising edge-triggered flip-flop clocked by MY_CLK)

```

```

Endpoint: status_o[UF]
          (output port clocked by MY_CLK)

```

```
Path Group: MY_CLK
```

```
Path Type: max
```

Des/Clust/Port	Wire Load Model	Library
fpnew_top	5K_hvrat1o_1_1	NangateOpenCellLibrary

Point	Incr	Path
--		

clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/mid_pipe_sum_q_reg[1][9]/CK (DFFR_X1)	0.00	0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/mid_pipe_sum_q_reg[1][9]/Q (DFFR_X1)	0.10	0.10 f
U1167/ZN (NOR4_X1)	0.14	0.25 r
U1248/ZN (OAI221_X1)	0.06	0.31 f
U1249/ZN (OAI211_X1)	0.04	0.35 r
U1250/ZN (AOI22_X1)	0.04	0.39 f
U1251/ZN (OAI211_X1)	0.04	0.43 r

U1252/ZN (INV_X1)	0.02	0.45 f
U1253/Z (MUX2_X1)	0.07	0.52 f
U1254/ZN (INV_X1)	0.03	0.55 r
U969/ZN (AND2_X1)	0.05	0.61 r
U1255/ZN (NAND2_X1)	0.03	0.64 f
U1257/ZN (NAND3_X1)	0.05	0.69 r
U1261/ZN (NAND2_X1)	0.03	0.72 f
U1263/ZN (NAND3_X1)	0.04	0.77 r
U1266/ZN (NAND2_X1)	0.03	0.80 f
U1268/ZN (NAND3_X1)	0.04	0.84 r
U1271/ZN (NAND2_X1)	0.03	0.87 f
U1273/ZN (NAND3_X1)	0.04	0.91 r
U1276/ZN (NAND2_X1)	0.03	0.94 f
U957/ZN (AND3_X1)	0.05	0.99 f
U1034/ZN (OAI21_X1)	0.07	1.06 r
U1279/ZN (NAND2_X1)	0.03	1.09 f
U958/ZN (AND3_X1)	0.05	1.14 f
U1162/ZN (OAI21_X1)	0.06	1.19 r
U1282/ZN (NAND2_X1)	0.03	1.23 f
U1284/ZN (NAND3_X1)	0.03	1.26 r
U1287/ZN (NAND2_X1)	0.03	1.29 f
U1288/ZN (OR2_X2)	0.08	1.37 f
U959/ZN (OAI222_X1)	0.10	1.47 r
U1037/ZN (INV_X1)	0.05	1.52 f
U1314/ZN (NAND2_X1)	0.05	1.57 r
U956/Z (BUF_X1)	0.10	1.67 r
U1354/ZN (OAI22_X1)	0.06	1.73 f
U1355/ZN (AOI211_X1)	0.12	1.85 r
U1356/ZN (OR2_X1)	0.05	1.90 r
U1360/ZN (NAND2_X1)	0.03	1.93 f
U1361/ZN (NOR2_X1)	0.04	1.97 r
U1362/ZN (NAND2_X1)	0.03	2.00 f
U1376/ZN (NAND2_X1)	0.03	2.03 r
U1409/ZN (INV_X1)	0.02	2.05 f
U1207/ZN (AND2_X2)	0.06	2.11 f
U1174/ZN (INV_X1)	0.07	2.18 r
U1062/ZN (AND2_X1)	0.05	2.23 r
U1149/ZN (OR2_X1)	0.03	2.26 r
U1418/ZN (AND2_X1)	0.04	2.30 r
U955/ZN (OR2_X1)	0.05	2.34 r
U1083/ZN (AND4_X1)	0.07	2.41 r
U1439/ZN (AOI21_X1)	0.06	2.48 f
U960/ZN (AND4_X1)	0.07	2.55 f
U1002/ZN (AND2_X1)	0.04	2.59 f
U1550/ZN (NOR2_X1)	0.04	2.63 r
U1551/ZN (OAI22_X1)	0.03	2.66 f
U1552/ZN (OAI21_X1)	0.04	2.70 r
U1553/ZN (NAND2_X1)	0.04	2.74 f
U1152/ZN (NOR2_X1)	0.07	2.81 r
U1565/ZN (NAND2_X1)	0.04	2.85 f
U1084/ZN (NOR2_X1)	0.06	2.92 r
U1587/ZN (NAND2_X1)	0.04	2.96 f
U1085/ZN (NOR2_X1)	0.06	3.02 r
U1609/ZN (NAND2_X1)	0.05	3.07 f
U1610/ZN (NOR2_X1)	0.06	3.12 r
U1613/ZN (AND2_X1)	0.06	3.18 r
U1614/ZN (NAND2_X1)	0.04	3.22 f
U1615/ZN (INV_X1)	0.03	3.25 r
U1215/ZN (AND2_X1)	0.05	3.30 r
U978/ZN (NAND2_X1)	0.04	3.34 f
U979/ZN (OR2_X1)	0.07	3.41 f
U1629/ZN (NAND2_X1)	0.04	3.45 r
U1630/ZN (INV_X1)	0.03	3.47 f
U1631/ZN (NAND4_X1)	0.03	3.51 r
U1632/ZN (NAND2_X1)	0.02	3.53 f
U961/ZN (AND2_X1)	0.05	3.58 f

U1635/ZN (INV_X1)	0.03	3.61 r
U1636/ZN (OAI21_X1)	0.04	3.64 f
U1224/ZN (AND4_X1)	0.06	3.70 f
status_o[UF] (out)	0.02	3.73 f
data arrival time		3.73
clock MY_CLK (rise edge)	4.30	4.30
clock network delay (ideal)	0.00	4.30
clock uncertainty	-0.07	4.23
output external delay	-0.50	3.73
data required time		3.73
--		

data required time		3.73
data arrival time		-3.73
--		

slack (MET)		0.00

.1.4 CSA multiplier

area

Report : area

Design : fpnew_top

Version: S-2021.06-SP4

Date : Mon Nov 17 17:02:05 2025

Library(s) Used:

NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:	458
Number of nets:	2617
Number of cells:	2128
Number of combinational cells:	1861
Number of sequential cells:	247
Number of macros/black boxes:	0
Number of buf/inv:	417
Number of references:	43

Combinational area:	2114.167996
Buf/Inv area:	232.750001
Noncombinational area:	1314.040042
Macro/Black Box area:	0.000000
Net Interconnect area:	undefined (Wire load has zero net area)

Total cell area:	3428.208039
Total area:	undefined

timing

Report : timing

-path full

-delay max

-max_paths 1

Design : fpnew_top

Version: S-2021.06-SP4

Date : Mon Nov 17 17:02:05 2025

Operating Conditions: typical Library: NangateOpenCellLibrary
Wire Load Model Mode: top

Startpoint: MY_CLK_r_REG142_S2
 (rising edge-triggered flip-flop clocked by MY_CLK)
Endpoint: status_o[NX]
 (output port clocked by MY_CLK)
Path Group: MY_CLK
Path Type: max

Des/Clust/Port	Wire Load Model	Library
-----	-----	-----
fpnew_top	5K_hvratio_1_1	NangateOpenCellLibrary

Point	Incr	Path
-----	-----	-----
clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
MY_CLK_r_REG142_S2/CK (DFFR_X1)	0.00	0.00 r
MY_CLK_r_REG142_S2/Q (DFFR_X1)	0.09	0.09 f
U945/ZN (OR2_X1)	0.06	0.14 f
U944/ZN (OR2_X1)	0.06	0.21 f
U970/ZN (XNOR2_X1)	0.06	0.27 f
U1632/ZN (NAND4_X1)	0.04	0.31 r
U1146/ZN (NAND2_X1)	0.03	0.34 f
U1633/ZN (NAND2_X1)	0.03	0.37 r
U1682/ZN (NAND2_X1)	0.03	0.41 f
U1684/ZN (OAI221_X1)	0.09	0.50 r
U848/ZN (INV_X1)	0.03	0.53 f
U921/ZN (AND2_X2)	0.05	0.57 f
U788/Z (BUF_X1)	0.06	0.63 f
U1732/ZN (AOI22_X1)	0.05	0.68 r
U1733/ZN (OAI221_X1)	0.05	0.74 f
U1734/ZN (INV_X1)	0.05	0.78 r
U1736/ZN (AOI22_X1)	0.03	0.81 f
U1737/ZN (OAI211_X1)	0.04	0.86 r
U880/ZN (OR2_X1)	0.04	0.90 r
U850/ZN (AND3_X1)	0.06	0.96 r
U961/ZN (NAND3_X1)	0.05	1.01 f
U876/Z (CLKBUF_X1)	0.05	1.05 f
U1851/ZN (OAI21_X1)	0.05	1.11 r
U1136/ZN (AND3_X1)	0.06	1.16 r
U1144/ZN (NAND2_X1)	0.02	1.19 f
U1852/ZN (OAI22_X1)	0.05	1.24 r
U980/ZN (AOI221_X1)	0.03	1.27 f
U801/ZN (NOR3_X1)	0.07	1.34 r
U1853/ZN (INV_X1)	0.04	1.38 f
U1889/ZN (OAI211_X1)	0.05	1.44 r
U812/ZN (AND4_X1)	0.07	1.51 r
U807/ZN (AND3_X1)	0.05	1.56 r
U954/ZN (AND2_X1)	0.04	1.60 r
U739/ZN (AND4_X2)	0.06	1.66 r
U1070/ZN (XNOR2_X1)	0.06	1.72 r
U1921/ZN (NAND4_X1)	0.04	1.76 f
U1922/ZN (OAI21_X1)	0.04	1.80 r
U1923/ZN (NAND2_X1)	0.03	1.83 f
U852/Z (BUF_X1)	0.04	1.87 f
U975/ZN (NAND2_X1)	0.03	1.91 r
status_o[NX] (out)	0.02	1.93 r
data arrival time		1.93
clock MY_CLK (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.07	2.43

resources

	Current	Set
Cell	Module	Implementation
add_i_root_add.i_root_gen_operation_groups[0].iopgroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/sub_287		
	DW01.add	parch (area, speed)
gen_operation_groups[0].iopgroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/sub_372		

.1.5 PPARCH multiplier

timing

```
Startpoint: MY_CLK_r_REG177_S2
            (rising edge-triggered flip-flop clocked by MY_CLK)
Endpoint: status_o[NX]
            (output port clocked by MY_CLK)
Path Group: MY_CLK
Path Type: max
```

Des/Clust/Port	Wire Load Model	Library	
fpnew_top	5K_hvratio_1_1	NangateOpenCellLibrary	
Point	Incr	Path	
clock MY_CLK (rise edge)	0.00	0.00	
clock network delay (ideal)	0.00	0.00	
MY_CLK_r_REG177_S2/CK (DFFR_X1)	0.00	0.00 r	
MY_CLK_r_REG177_S2/Q (DFFR_X1)	0.09	0.09 f	
U945/ZN (OR2_X1)	0.06	0.14 f	
U944/ZN (OR2_X1)	0.06	0.21 f	
U970/ZN (XNOR2_X1)	0.06	0.27 f	
U1632/ZN (NAND4_X1)	0.04	0.31 r	
U1146/ZN (NAND2_X1)	0.03	0.34 f	
U1633/ZN (NAND2_X1)	0.03	0.37 r	
U1682/ZN (NAND2_X1)	0.03	0.40 f	
U1684/ZN (OAI221_X1)	0.09	0.50 r	
U848/ZN (INV_X1)	0.02	0.52 f	
U921/ZN (AND2_X1)	0.05	0.57 f	
U1705/ZN (AOI22_X1)	0.05	0.62 r	
U1706/ZN (OAI221_X1)	0.05	0.67 f	
U1707/ZN (INV_X1)	0.03	0.70 r	
U825/ZN (AND2_X1)	0.05	0.76 r	
U1728/ZN (NOR4_X1)	0.03	0.79 f	
U1016/ZN (AOI211_X1)	0.08	0.86 r	
U1133/ZN (OAI211_X1)	0.05	0.91 f	
U1748/ZN (INV_X1)	0.04	0.95 r	
U1770/ZN (NAND2_X1)	0.04	0.99 f	
U984/ZN (INV_X1)	0.04	1.03 r	
U1850/ZN (AOI22_X1)	0.03	1.07 f	
U1851/ZN (OAI21_X1)	0.04	1.11 r	
U1136/ZN (AND3_X1)	0.06	1.16 r	
U1144/ZN (NAND2_X1)	0.02	1.19 f	
U1852/ZN (OAI22_X1)	0.05	1.24 r	
U980/ZN (AOI221_X1)	0.03	1.27 f	
U801/ZN (NOR3_X1)	0.07	1.34 r	
U1853/ZN (INV_X1)	0.04	1.38 f	
U1889/ZN (OAI211_X1)	0.05	1.43 r	
U812/ZN (AND4_X1)	0.07	1.51 r	
U807/ZN (AND3_X1)	0.05	1.56 r	
U954/ZN (AND2_X1)	0.04	1.60 r	
U739/ZN (AND4_X2)	0.06	1.66 r	
U1070/ZN (XNOR2_X1)	0.06	1.72 r	
U1921/ZN (NAND4_X1)	0.04	1.76 f	
U1922/ZN (OAI21_X1)	0.04	1.80 r	
U1923/ZN (NAND2_X1)	0.03	1.83 f	
U852/Z (BUF_X1)	0.04	1.87 f	
U975/ZN (NAND2_X1)	0.03	1.90 r	
status_o[NX] (out)	0.02	1.93 r	
data arrival time		1.93	
clock MY_CLK (rise edge)	2.50	2.50	
clock network delay (ideal)	0.00	2.50	
clock uncertainty	-0.07	2.43	
output external delay	-0.50	1.93	
data required time		1.93	
data required time		1.93	
data arrival time		-1.93	
slack (MET)		0.00	

Resource Sharing Report for design fpnew_top in file ./codes/fpnew_top.sv

Implementation Report

Cell	Module	Current Implementation	Set Implementation
add_1_root_add_1_root_gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/sub_287	DW01_add	pparch (area, speed)	
gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/sub_372	DW01_sub	pparch (speed)	
add_1_root_gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/add_368_2	DW01_add	pparch (speed)	
gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/sub_293	DW01_sub	pparch (area, speed)	
add_0_root_gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/add_515	DW01_inc	rpl	
gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/mult_325	DW02_mult	pparch (area)	pparch
		mult_arch: and	
gen_operation_groups[0].iogroup_block/gen_parallel_slices[4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.i_fma/add_542	DW01_inc	rpl	

.2 R4-MBE multiplier implementation

.2.1 compile

area

```
*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Thu Nov 27 17:40:45 2025
*****

Library(s) Used:

      NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
      NangateOpenCellLibrary_typical_ecsm.db)

Number of ports:                1914
Number of nets:                 4193
Number of cells:                2353
Number of combinational cells:  2135
Number of sequential cells:     160
Number of macros/black boxes:   0
Number of buf/inv:              396
Number of references:           15

Combinational area:             2639.252031
Buf/Inv area:                   218.918002
Noncombinational area:          853.860027
Macro/Black Box area:           0.000000
Net Interconnect area:          undefined (Wire load has zero net area)

Total cell area:                 3493.112058
Total area:                      undefined
```

timing

```
*****
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Thu Nov 27 17:37:51 2025
*****

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
            [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
            lane_instance.i_fma/mid_pipe_sum_q_reg[1][7]
            (rising edge-triggered flip-flop clocked by MY_CLK)
Endpoint:  status_o[UF]
            (output port clocked by MY_CLK)
Path Group: MY_CLK
Path Type: max

Des/Clust/Port      Wire Load Model      Library
-----
fpnew_top           5K_hvratio_1_1       NangateOpenCellLibrary

Point                                     Incr                                     Path
```

```

--
-----

clock MY_CLK (rise edge)                                0.00      0.00
clock network delay (ideal)                              0.00      0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/mid_pipe_sum_q_reg[1][7]/CK (DFFR_X1)              0.00      0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/mid_pipe_sum_q_reg[1][7]/Q (DFFR_X1)              0.10      0.10 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/in_i[7] (lzc_00000013_1)                   0.00      0.10 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/U29/ZN (INV_X1)                            0.04      0.14 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/U4/ZN (AND4_X2)                            0.08      0.22 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/U52/ZN (NAND4_X2)                          0.07      0.29 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/U44/ZN (NAND2_X1)                          0.04      0.33 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/U43/ZN (NAND2_X1)                          0.03      0.37 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/i_lzc/cnt_o[2] (lzc_00000013_1)                  0.00      0.37 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/I2[2] (
  fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__add_512_DP_OP_405_6790_1
  )
  )                                                        0.00      0.37 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/U172/ZN (INV_X1)       0.03      0.40 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/U190/ZN (NAND3_X1)     0.03      0.43 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/U182/ZN (AND2_X1)     0.04      0.47 f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/U184/ZN (NAND2_X1)    0.03      0.49 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
  active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
  i_fma/add_512_DP_OP_405_6790_49/U187/ZN (NAND3_X1)

```

	0.03	0.52	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/add_512_DP_OP_405_6790_49/U189/ZN (NAND3_X1)			
	0.03	0.56	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/add_512_DP_OP_405_6790_49/O20 (			
fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__add_512_DP_OP_405_6790_1			
)			
	0.00	0.56	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U413/ZN (NAND2_X1)			
	0.04	0.59	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U369/ZN (OR2_X1)			
	0.06	0.66	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U427/ZN (AND3_X1)			
	0.04	0.70	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U10/ZN (AND2_X2)			
	0.04	0.75	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U359/Z (BUF_X2)			
	0.05	0.80	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U356/ZN (NAND2_X1)			
	0.03	0.83	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U278/ZN (NAND4_X1)			
	0.05	0.89	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U416/ZN (NAND2_X1)			
	0.04	0.93	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U185/ZN (AND3_X2)			
	0.05	0.98	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U421/ZN (NAND4_X2)			
	0.06	1.04	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U289/ZN (OR3_X2)			
	0.09	1.13	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U1264/ZN (OAI21_X1)			
	0.06	1.19	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U327/ZN (OR2_X1)			
	0.04	1.23	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U454/ZN (NAND2_X1)			

	0.03	1.26	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U1189/ZN (NAND2_X1)			
	0.03	1.28	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U1216/ZN (AND2_X1)			
	0.05	1.33	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U181/ZN (AND4_X1)			
	0.06	1.39	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U1137/ZN (OAI211_X1)			
	0.04	1.43	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U405/Z (BUF_X2)			
	0.05	1.48	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U1254/ZN (NAND4_X1)			
	0.05	1.53	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/abs_value_i[2] (fpnew_rounding_0000000f)			
	0.00	1.53	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/U8/ZN (NAND2_X1)			
	0.03	1.57	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/U7/ZN (NOR2_X1)			
	0.05	1.61	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/U49/ZN (AND4_X1)			
	0.07	1.68	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/U70/ZN (NAND2_X1)			
	0.03	1.71	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/U71/ZN (XNOR2_X1)			
	0.07	1.78	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/i_fpnew_rounding/abs_rounded_o[9] (fpnew_rounding_0000000f)			
	0.00	1.78	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U339/ZN (AND2_X1)			
	0.05	1.83	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U433/ZN (NAND4_X1)			
	0.05	1.88	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.			
i_fma/U435/ZN (NAND2_X1)			
	0.04	1.91	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].			


```

    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/U1204/ZN (AND2_X1)
                                0.04      1.95 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.
    i_fma/status_o[UF] (
    fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB_)
                                0.00      1.95 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/U35/ZN (AND2_X1)
                                0.04      1.99 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].
    active_format.i_fmt_slice/status_o[UF] (
    fpnew_opgroup_fmt_slice_0_4_00000010_0_00000002_3__logic_Z_1yB_)
                                0.00      1.99 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (
    rr_arb_tree_00000005_1_024242)
                                0.00      1.99 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/U78/ZN (NAND2_X1)
                                0.02      2.01 f
gen_operation_groups[0].i_opgroup_block/i_arbiter/U77/ZN (NAND4_X1)
                                0.03      2.05 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (
    rr_arb_tree_00000005_1_024242)
                                0.00      2.05 r
gen_operation_groups[0].i_opgroup_block/status_o[UF] (
    fpnew_opgroup_block_0_00000010_0_01_4_155_3__logic_Z_1yB__395949)
                                0.00      2.05 r
i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00      2.05 r
i_arbiter/U42/ZN (NAND2_X1)
                                0.03      2.07 f
i_arbiter/U41/ZN (NAND2_X1)
                                0.03      2.11 r
i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00      2.11 r
status_o[UF] (out)
                                0.02      2.13 r
data arrival time
                                2.13

clock MY_CLK (rise edge)
                                2.70      2.70
clock network delay (ideal)
                                0.00      2.70
clock uncertainty
                                -0.07     2.63
output external delay
                                -0.50     2.13
data required time
                                2.13

--
-----

data required time
                                2.13
data arrival time
                                -2.13
--
-----

slack (MET)
                                0.00

```

.2.2 compile and optimize_registers

area

```

*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Thu Nov 27 17:11:55 2025
*****

```

Library(s) Used:

NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/

NangateOpenCellLibrary_typical_ecsm.db)

```

Number of ports:                1853
Number of nets:                 2901
Number of cells:                1844
Number of combinational cells:  1448
Number of sequential cells:     158
Number of macros/black boxes:   0
Number of buf/inv:              242
Number of references:           13

Combinational area:             1911.210017
Buf/Inv area:                   134.596001
Noncombinational area:          840.560027
Macro/Black Box area:           0.000000
Net Interconnect area:          undefined (Wire load has zero net area)

Total cell area:                 2751.770044
Total area:                      undefined

```

timing

```
*****
```

```
Report : timing
```

```

-path full
-delay max
-max_paths 1

```

```
Design : fpnew_top
```

```
Version: S-2021.06-SP4
```

```
Date : Thu Nov 27 17:09:28 2025
```

```
*****
```

```
Operating Conditions: typical Library: NangateOpenCellLibrary
```

```
Wire Load Model Mode: top
```

```

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
             [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
             lane_instance.i_fma/MY_CLK_r_REG51_S2
             (rising edge-triggered flip-flop clocked by MY_CLK)

```

```

Endpoint: status_o[UF]
          (output port clocked by MY_CLK)

```

```
Path Group: MY_CLK
```

```
Path Type: max
```

Des/Clust/Port	Wire Load Model	Library
fpnew_top	5K_hvrat1o_1_1	NangateOpenCellLibrary

Point	Incr	Path
clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/MY_CLK_r_REG51_S2/CK (DFFS_X2)	0.00	0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/MY_CLK_r_REG51_S2/Q (DFFS_X2)	0.14	0.14 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U267/ZN (AND2_X1)		

gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U268/ZN (AND2_X2)	0.06	0.19	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U9/Z (BUF_X1)	0.05	0.24	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U1323/ZN (NAND2_X1)	0.04	0.28	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U134/ZN (NAND2_X1)	0.03	0.31	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/A[0] (fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_inc_0)	0.05	0.36	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U9/CO (HA_X1)	0.00	0.36	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U18/ZN (AND2_X1)	0.07	0.43	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U7/CO (HA_X1)	0.05	0.48	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U6/CO (HA_X1)	0.06	0.54	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U5/CO (HA_X1)	0.06	0.59	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U4/CO (HA_X1)	0.06	0.65	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U15/ZN (NAND2_X1)	0.06	0.71	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/U14/ZN (XNOR2_X1)	0.03	0.74	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/add_542/SUM[9] (fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB__DW01_inc_0)	0.05	0.79	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U449/ZN (NAND2_X1)	0.00	0.79	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U349/ZN (NAND2_X1)	0.03	0.82	r

gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U438/ZN (NOR2_X1)	0.03	0.85	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U1260/ZN (NAND4_X1)	0.04	0.89	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U266/ZN (AND2_X2)	0.05	0.94	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U1295/ZN (NAND4_X1)	0.06	1.00	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/abs_value_i[6] (fpnew_rounding_0000000f)	0.05	1.06	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U53/ZN (AND2_X2)	0.00	1.06	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U8/ZN (AND4_X1)	0.06	1.12	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U48/ZN (AND3_X1)	0.08	1.20	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/U75/ZN (XNOR2_X1)	0.06	1.25	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/i_fpnew_rounding/abs_rounded_o[12] (fpnew_rounding_0000000f)	0.06	1.31	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U309/ZN (AND4_X1)	0.00	1.31	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U456/ZN (NAND3_X1)	0.07	1.38	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U323/ZN (NAND2_X1)	0.04	1.42	f
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/U1292/ZN (AND2_X1)	0.03	1.45	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance. i_fma/status_o[UF] (fpnew_fma_4_00000002_3__logic_Z_1yB__logic_Z_1yB_)	0.04	1.49	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4]. active_format.i_fmt_slice/U36/ZN (AND2_X1)	0.00	1.49	r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].	0.04	1.52	r

```

    active_format.i_fmt_slice/status_o[UF] (
    fpnew_opgroup_fmt_slice_0_4_00000010_0_00000002_3__logic_Z_1yB_)
                                0.00      1.52 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_i[4][status][UF] (
    rr_arb_tree_00000005_1_024242)
                                0.00      1.52 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/U31/ZN (AND2_X1)
                                0.04      1.56 r
gen_operation_groups[0].i_opgroup_block/i_arbiter/data_o[status][UF] (
    rr_arb_tree_00000005_1_024242)
                                0.00      1.56 r
gen_operation_groups[0].i_opgroup_block/status_o[UF] (
    fpnew_opgroup_block_0_00000010_0_01_4_155_3__logic_Z_1yB__395949)
                                0.00      1.56 r
i_arbiter/data_i[0][status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00      1.56 r
i_arbiter/U27/ZN (AND2_X1)
                                0.04      1.61 r
i_arbiter/data_o[status][UF] (rr_arb_tree_00000004_1_891242)
                                0.00      1.61 r
status_o[UF] (out)
                                0.02      1.63 r
data arrival time
                                1.63

    clock MY_CLK (rise edge)
                                2.20      2.20
    clock network delay (ideal)
                                0.00      2.20
    clock uncertainty
                                -0.07     2.13
    output external delay
                                -0.50     1.63
    data required time
                                1.63

--
-----

data required time
                                1.63
data arrival time
                                -1.63
--
-----

slack (MET)
                                0.00

```

.2.3 compile_ultra

area

```

*****
Report : area
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Thu Nov 27 17:48:41 2025
*****

```

```

Information: Updating design information... (UID-85)
Library(s) Used:

```

```

    NangateOpenCellLibrary (File: /eda/dk/nangate45/synopsys/
    NangateOpenCellLibrary_typical_ecsm.db)

```

```

Number of ports:          96
Number of nets:          1661
Number of cells:          1434
Number of combinational cells: 1291
Number of sequential cells: 141
Number of macros/black boxes: 0
Number of buf/inv:        179
Number of references:      37

```

```

Combinational area:      1657.712004
Buf/Inv area:            96.824001
Noncombinational area:   752.780024

```

```

Macro/Black Box area:          0.000000
Net Interconnect area:        undefined (Wire load has zero net area)

Total cell area:              2410.492028
Total area:                   undefined

```

timing

```

*****
Report : timing
        -path full
        -delay max
        -max_paths 1
Design : fpnew_top
Version: S-2021.06-SP4
Date   : Thu Nov 27 17:47:34 2025
*****

```

```

Operating Conditions: typical   Library: NangateOpenCellLibrary
Wire Load Model Mode: top

```

```

Startpoint: gen_operation_groups[0].i_opgroup_block/gen_parallel_slices
             [4].active_format.i_fmt_slice/gen_num_lanes[0].active_lane.
             lane_instance.i_fma/mid_pipe_sum_q_reg[1][18]
             (rising edge-triggered flip-flop clocked by MY_CLK)
Endpoint:   status_o[UF]
             (output port clocked by MY_CLK)
Path Group: MY_CLK
Path Type:  max

```

Des/Clust/Port	Wire Load Model	Library
fpnew_top	5K_hvratio_1_1	NangateOpenCellLibrary

Point	Incr	Path
--		

clock MY_CLK (rise edge)	0.00	0.00
clock network delay (ideal)	0.00	0.00
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].		
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.		
i_fma/mid_pipe_sum_q_reg[1][18]/CK (DFFR_X1)	0.00	0.00 r
gen_operation_groups[0].i_opgroup_block/gen_parallel_slices[4].		
active_format.i_fmt_slice/gen_num_lanes[0].active_lane.lane_instance.		
i_fma/mid_pipe_sum_q_reg[1][18]/Q (DFFR_X1)	0.09	0.09 f
U1974/ZN (NOR2_X2)	0.06	0.15 r
U2009/ZN (AND4_X2)	0.08	0.23 r
U1892/ZN (AND2_X1)	0.05	0.28 r
U1927/ZN (NAND2_X1)	0.02	0.30 f
U1962/ZN (AND2_X2)	0.04	0.35 f
U2041/ZN (NOR2_X2)	0.05	0.40 r
U2606/ZN (NOR2_X1)	0.03	0.43 f
U2608/ZN (AOI21_X1)	0.05	0.48 r
U2119/ZN (NAND2_X1)	0.03	0.51 f
U2118/ZN (NAND2_X1)	0.03	0.54 r
U1872/ZN (NAND2_X1)	0.04	0.58 f
U1905/ZN (OAI211_X1)	0.09	0.67 r
U2634/ZN (NOR2_X1)	0.04	0.71 f
U1054/Z (BUF_X2)	0.05	0.76 f
U2656/ZN (AOI22_X1)	0.07	0.83 r
U2659/ZN (AND2_X1)	0.05	0.88 r
U2660/ZN (OR2_X1)	0.04	0.92 r

U2665/ZN (OAI211_X1)	0.04	0.96 f
U2064/ZN (OAI211_X1)	0.06	1.01 r
U974/ZN (AND3_X1)	0.07	1.09 r
U2717/ZN (INV_X1)	0.03	1.12 f
U1081/ZN (NAND2_X1)	0.06	1.18 r
U2736/ZN (OAI21_X1)	0.04	1.22 f
U2243/ZN (AND2_X1)	0.04	1.26 f
U2737/ZN (INV_X1)	0.03	1.29 r
U2749/ZN (NAND4_X1)	0.05	1.33 f
U2127/ZN (OAI21_X2)	0.07	1.40 r
U2077/ZN (INV_X1)	0.03	1.43 f
U2035/ZN (INV_X1)	0.03	1.46 r
U1866/ZN (OR2_X1)	0.04	1.51 r
U1856/ZN (AND3_X2)	0.08	1.58 r
U1865/ZN (AND2_X1)	0.06	1.64 r
U2087/ZN (NAND4_X1)	0.04	1.68 f
U2036/ZN (NAND2_X1)	0.04	1.72 r
U2007/ZN (AND2_X1)	0.04	1.77 r
U2095/ZN (NAND4_X1)	0.05	1.81 f
U2081/ZN (NAND2_X1)	0.04	1.85 r
U1694/ZN (AND3_X1)	0.06	1.90 r
status_o[UF] (out)	0.02	1.93 r
data arrival time		1.93
clock MY_CLK (rise edge)	2.50	2.50
clock network delay (ideal)	0.00	2.50
clock uncertainty	-0.07	2.43
output external delay	-0.50	1.93
data required time		1.93
--		

data required time		1.93
data arrival time		-1.93
--		

slack (MET)		0.00

.3 Testbench

In addition to the reports listed before, here is shown the complete output of the testbench made for the multiplier. The A column and the B one show the multiplying and the multiplier, M column shows the results obtained through the circuit, Ideal_M the ideal results, Error the difference between the actual value and the ideal one.

A: x		B: 0		M: 0		Ideal_M: x		Error: x
A:199		B:231		M:45969		Ideal_M:45969		Error: 0
A: 78		B:118		M: 9204		Ideal_M: 9204		Error: 0
A: 77		B: 58		M: 4466		Ideal_M: 4466		Error: 0
A:188		B: 69		M:12972		Ideal_M:12972		Error: 0
A: 80		B:230		M:18400		Ideal_M:18400		Error: 0
A:249		B:236		M:58764		Ideal_M:58764		Error: 0
A: 60		B: 71		M: 4260		Ideal_M: 4260		Error: 0
A:150		B: 54		M: 8100		Ideal_M: 8100		Error: 0
A:104		B: 83		M: 8632		Ideal_M: 8632		Error: 0
A:178		B:233		M:41474		Ideal_M:41474		Error: 0
A:182		B:219		M:39858		Ideal_M:39858		Error: 0
A: 25		B:180		M: 4500		Ideal_M: 4500		Error: 0
A: 39		B: 0		M: 0		Ideal_M: 0		Error: 0
A:120		B:144		M:17280		Ideal_M:17280		Error: 0
A: 25		B: 7		M: 175		Ideal_M: 175		Error: 0
A: 26		B:186		M: 4836		Ideal_M: 4836		Error: 0
A: 36		B:240		M: 8640		Ideal_M: 8640		Error: 0
A: 17		B: 80		M: 1360		Ideal_M: 1360		Error: 0

A: 98		B:248		M:24304		Ideal_M:24304		Error:	0
A: 29		B:234		M: 6786		Ideal_M: 6786		Error:	0
A: 4		B:231		M: 924		Ideal_M: 924		Error:	0
A:218		B:241		M:52538		Ideal_M:52538		Error:	0
A:116		B: 90		M:10440		Ideal_M:10440		Error:	0
A:109		B:172		M:18748		Ideal_M:18748		Error:	0
A:131		B:213		M:27903		Ideal_M:27903		Error:	0
A:106		B: 89		M: 9434		Ideal_M: 9434		Error:	0
A:190		B: 0		M: 0		Ideal_M: 0		Error:	0
A: 34		B:215		M: 7310		Ideal_M: 7310		Error:	0
A:240		B:231		M:55440		Ideal_M:55440		Error:	0
A: 35		B:156		M: 5460		Ideal_M: 5460		Error:	0
A:241		B:231		M:55671		Ideal_M:55671		Error:	0
A:234		B: 92		M:21528		Ideal_M:21528		Error:	0
A:216		B:200		M:43200		Ideal_M:43200		Error:	0
A: 42		B:162		M: 6804		Ideal_M: 6804		Error:	0
A:140		B: 99		M:13860		Ideal_M:13860		Error:	0
A: 64		B:122		M: 7808		Ideal_M: 7808		Error:	0
A:220		B: 5		M: 1100		Ideal_M: 1100		Error:	0
A: 16		B: 4		M: 64		Ideal_M: 64		Error:	0
A: 15		B: 3		M: 45		Ideal_M: 45		Error:	0
A:235		B: 77		M:18095		Ideal_M:18095		Error:	0
A: 91		B:208		M:18928		Ideal_M:18928		Error:	0
A:173		B:117		M:20241		Ideal_M:20241		Error:	0
A:183		B: 26		M: 4758		Ideal_M: 4758		Error:	0
A:114		B:141		M:16074		Ideal_M:16074		Error:	0
A: 20		B:176		M: 3520		Ideal_M: 3520		Error:	0
A:183		B: 14		M: 2562		Ideal_M: 2562		Error:	0
A:131		B: 26		M: 3406		Ideal_M: 3406		Error:	0
A: 4		B: 40		M: 160		Ideal_M: 160		Error:	0
A: 61		B: 78		M: 4758		Ideal_M: 4758		Error:	0
A:209		B: 63		M:13167		Ideal_M:13167		Error:	0
A: 13		B: 61		M: 793		Ideal_M: 793		Error:	0
A: 20		B:196		M: 3920		Ideal_M: 3920		Error:	0
A:188		B:225		M:42300		Ideal_M:42300		Error:	0
A: 84		B: 92		M: 7728		Ideal_M: 7728		Error:	0
A:148		B: 71		M:10508		Ideal_M:10508		Error:	0
A: 78		B:241		M:18798		Ideal_M:18798		Error:	0
A: 3		B:144		M: 432		Ideal_M: 432		Error:	0
A:162		B:182		M:29484		Ideal_M:29484		Error:	0
A: 62		B: 36		M: 2232		Ideal_M: 2232		Error:	0
A: 98		B:215		M:21070		Ideal_M:21070		Error:	0
A:103		B:164		M:16892		Ideal_M:16892		Error:	0
A:136		B:192		M:26112		Ideal_M:26112		Error:	0
A:233		B: 55		M:12815		Ideal_M:12815		Error:	0
A:131		B:174		M:22794		Ideal_M:22794		Error:	0
A: 90		B:116		M:10440		Ideal_M:10440		Error:	0
A:216		B:201		M:43416		Ideal_M:43416		Error:	0
A:197		B:218		M:42946		Ideal_M:42946		Error:	0
A: 20		B: 0		M: 0		Ideal_M: 0		Error:	0
A:198		B:155		M:30690		Ideal_M:30690		Error:	0
A:232		B:112		M:25984		Ideal_M:25984		Error:	0
A:124		B:122		M:15128		Ideal_M:15128		Error:	0
A:222		B:101		M:22422		Ideal_M:22422		Error:	0
A:170		B: 97		M:16490		Ideal_M:16490		Error:	0
A: 31		B:159		M: 4929		Ideal_M: 4929		Error:	0
A: 62		B: 89		M: 5518		Ideal_M: 5518		Error:	0
A: 55		B:237		M:13035		Ideal_M:13035		Error:	0
A:224		B: 9		M: 2016		Ideal_M: 2016		Error:	0
A:165		B:121		M:19965		Ideal_M:19965		Error:	0
A:141		B: 69		M: 9729		Ideal_M: 9729		Error:	0
A: 46		B:233		M:10718		Ideal_M:10718		Error:	0
A:154		B:225		M:34650		Ideal_M:34650		Error:	0
A:184		B: 49		M: 9016		Ideal_M: 9016		Error:	0
A:191		B:115		M:21965		Ideal_M:21965		Error:	0
A: 77		B:204		M:15708		Ideal_M:15708		Error:	0

A:238		B: 99		M:23562		Ideal_M:23562		Error:	0
A:213		B: 33		M: 7029		Ideal_M: 7029		Error:	0
A:213		B:117		M:24921		Ideal_M:24921		Error:	0
A: 8		B:142		M: 1136		Ideal_M: 1136		Error:	0
A: 83		B: 36		M: 2988		Ideal_M: 2988		Error:	0
A:239		B:192		M:45888		Ideal_M:45888		Error:	0
A: 18		B:131		M: 2358		Ideal_M: 2358		Error:	0
A:226		B: 79		M:17854		Ideal_M:17854		Error:	0
A:198		B: 26		M: 5148		Ideal_M: 5148		Error:	0
A:169		B: 68		M:11492		Ideal_M:11492		Error:	0
A:176		B:109		M:19184		Ideal_M:19184		Error:	0
A: 41		B: 85		M: 3485		Ideal_M: 3485		Error:	0
A:107		B: 91		M: 9737		Ideal_M: 9737		Error:	0
A:162		B: 5		M: 810		Ideal_M: 810		Error:	0
A:198		B:133		M:26334		Ideal_M:26334		Error:	0
A:153		B: 77		M:11781		Ideal_M:11781		Error:	0
A:103		B: 68		M: 7004		Ideal_M: 7004		Error:	0
A:129		B:114		M:14706		Ideal_M:14706		Error:	0
A:136		B: 49		M: 6664		Ideal_M: 6664		Error:	0
A: 62		B: 56		M: 3472		Ideal_M: 3472		Error:	0
A: 88		B:170		M:14960		Ideal_M:14960		Error:	0
A: 14		B: 87		M: 1218		Ideal_M: 1218		Error:	0
A:240		B: 83		M:19920		Ideal_M:19920		Error:	0
A: 9		B: 23		M: 207		Ideal_M: 207		Error:	0
A:248		B: 47		M:11656		Ideal_M:11656		Error:	0
A:156		B: 43		M: 6708		Ideal_M: 6708		Error:	0
A:213		B:191		M:40683		Ideal_M:40683		Error:	0
A:196		B: 60		M:11760		Ideal_M:11760		Error:	0
A: 76		B:237		M:18012		Ideal_M:18012		Error:	0
A:143		B: 82		M:11726		Ideal_M:11726		Error:	0
A:122		B:228		M:27816		Ideal_M:27816		Error:	0
A: 28		B:179		M: 5012		Ideal_M: 5012		Error:	0
A:254		B:150		M:38100		Ideal_M:38100		Error:	0
A:165		B: 4		M: 660		Ideal_M: 660		Error:	0
A:108		B:238		M:25704		Ideal_M:25704		Error:	0
A: 61		B:138		M: 8418		Ideal_M: 8418		Error:	0
A:135		B:246		M:33210		Ideal_M:33210		Error:	0
A:148		B: 7		M: 1036		Ideal_M: 1036		Error:	0
A:234		B: 87		M:20358		Ideal_M:20358		Error:	0
A: 40		B:188		M: 7520		Ideal_M: 7520		Error:	0
A:254		B:247		M:62738		Ideal_M:62738		Error:	0
A:238		B:204		M:48552		Ideal_M:48552		Error:	0
A: 37		B: 65		M: 2405		Ideal_M: 2405		Error:	0
A:198		B: 81		M:16038		Ideal_M:16038		Error:	0
A:189		B:177		M:33453		Ideal_M:33453		Error:	0
A:192		B:104		M:19968		Ideal_M:19968		Error:	0
A:162		B:156		M:25272		Ideal_M:25272		Error:	0
A:251		B:240		M:60240		Ideal_M:60240		Error:	0
A: 84		B:158		M:13272		Ideal_M:13272		Error:	0
A: 72		B:178		M:12816		Ideal_M:12816		Error:	0
A: 93		B:184		M:17112		Ideal_M:17112		Error:	0
A: 65		B:128		M: 8320		Ideal_M: 8320		Error:	0
A:124		B:204		M:25296		Ideal_M:25296		Error:	0
A:158		B:197		M:31126		Ideal_M:31126		Error:	0
A:104		B: 72		M: 7488		Ideal_M: 7488		Error:	0
A:104		B: 55		M: 5720		Ideal_M: 5720		Error:	0
A:248		B: 9		M: 2232		Ideal_M: 2232		Error:	0
A: 67		B: 80		M: 5360		Ideal_M: 5360		Error:	0
A:118		B:206		M:24308		Ideal_M:24308		Error:	0
A:104		B:228		M:23712		Ideal_M:23712		Error:	0
A: 28		B:103		M: 2884		Ideal_M: 2884		Error:	0
A:183		B: 90		M:16470		Ideal_M:16470		Error:	0
A:127		B:175		M:22225		Ideal_M:22225		Error:	0
A:218		B: 26		M: 5668		Ideal_M: 5668		Error:	0
A: 88		B:211		M:18568		Ideal_M:18568		Error:	0
A:228		B: 74		M:16872		Ideal_M:16872		Error:	0

A:194		B: 72		M:13968		Ideal_M:13968		Error:	0
A:116		B:181		M:20996		Ideal_M:20996		Error:	0
A:219		B: 25		M: 5475		Ideal_M: 5475		Error:	0
A: 67		B:221		M:14807		Ideal_M:14807		Error:	0
A: 99		B: 52		M: 5148		Ideal_M: 5148		Error:	0
A:162		B: 94		M:15228		Ideal_M:15228		Error:	0
A:201		B: 60		M:12060		Ideal_M:12060		Error:	0
A: 62		B: 91		M: 5642		Ideal_M: 5642		Error:	0
A: 40		B:246		M: 9840		Ideal_M: 9840		Error:	0
A:191		B: 51		M: 9741		Ideal_M: 9741		Error:	0
A: 53		B:250		M:13250		Ideal_M:13250		Error:	0
A:159		B:118		M:18762		Ideal_M:18762		Error:	0
A: 60		B: 79		M: 4740		Ideal_M: 4740		Error:	0
A: 51		B:214		M:10914		Ideal_M:10914		Error:	0
A: 7		B: 21		M: 147		Ideal_M: 147		Error:	0
A: 40		B:184		M: 7360		Ideal_M: 7360		Error:	0
A:188		B:187		M:35156		Ideal_M:35156		Error:	0
A: 4		B:247		M: 988		Ideal_M: 988		Error:	0
A: 14		B:248		M: 3472		Ideal_M: 3472		Error:	0
A: 15		B:130		M: 1950		Ideal_M: 1950		Error:	0
A: 3		B: 13		M: 39		Ideal_M: 39		Error:	0
A: 59		B:218		M:12862		Ideal_M:12862		Error:	0
A:253		B: 63		M:15939		Ideal_M:15939		Error:	0
A:190		B:197		M:37430		Ideal_M:37430		Error:	0
A: 42		B: 31		M: 1302		Ideal_M: 1302		Error:	0
A:230		B:129		M:29670		Ideal_M:29670		Error:	0
A:251		B:113		M:28363		Ideal_M:28363		Error:	0
A:157		B: 48		M: 7536		Ideal_M: 7536		Error:	0
A:246		B: 2		M: 492		Ideal_M: 492		Error:	0
A:111		B:241		M:26751		Ideal_M:26751		Error:	0
A: 50		B:100		M: 5000		Ideal_M: 5000		Error:	0
A:141		B:216		M:30456		Ideal_M:30456		Error:	0
A:101		B: 67		M: 6767		Ideal_M: 6767		Error:	0
A:196		B:101		M:19796		Ideal_M:19796		Error:	0
A: 96		B:240		M:23040		Ideal_M:23040		Error:	0
A:175		B: 59		M:10325		Ideal_M:10325		Error:	0
A:191		B: 80		M:15280		Ideal_M:15280		Error:	0
A: 46		B:145		M: 6670		Ideal_M: 6670		Error:	0
A: 16		B:157		M: 2512		Ideal_M: 2512		Error:	0
A: 4		B:153		M: 612		Ideal_M: 612		Error:	0
A:180		B:164		M:29520		Ideal_M:29520		Error:	0
A: 7		B:110		M: 770		Ideal_M: 770		Error:	0
A:248		B:197		M:48856		Ideal_M:48856		Error:	0
A: 47		B:186		M: 8742		Ideal_M: 8742		Error:	0
A:150		B: 25		M: 3750		Ideal_M: 3750		Error:	0
A: 16		B:130		M: 2080		Ideal_M: 2080		Error:	0
A: 77		B:101		M: 7777		Ideal_M: 7777		Error:	0
A: 54		B: 2		M: 108		Ideal_M: 108		Error:	0
A: 57		B:160		M: 9120		Ideal_M: 9120		Error:	0